



**1** Evaluate the following, given the functions:

$$f(x) = 4x^3$$

$$g(x) = x + 5$$

**a**  $f(-2)$

**b**  $g(f(-2))$

**c**  $f(g(0))$

**2** Evaluate  $f(g(2))$ , given the functions:

$$f(x) = 4x - 10$$

$$g(x) = 3 + \frac{3}{x}$$

3 Consider the following functions:

$$f(x) = -2x - 6$$

$$g(x) = 5x - 7$$

**a** Find  $f(7)$ .

**b** Hence, evaluate  $g(f(7))$ .

**c** Find  $g(7)$ .

**d** Hence, evaluate  $f(g(7))$ .

e Does  $f(g(x)) = g(f(x))$  for all  $x$ ?

**4** Find the composite function  $f(g(x))$  for the following:

**a**  $f(x) = x^3$   
 $g(x) = 9x - 4$

**b**  $f(x) = x^2$   
 $g(x) = x^2 + 1$

**c**  $f(x) = \sqrt{x}$   
 $g(x) = 4x - 3$

**d**  $f(x) = 4x + 9$   
 $g(x) = x^3$

e  $f(x) = 4x - 3$   
 $g(x) = x^2$

**5** Evaluate the following, given the functions:

$$f(x) = -2x + 2$$

$$g(x) = 4x^2 - 8$$

$$r(x) = -3x - 8$$

**a**  $g(6)$

**b**  $f(g(6))$

**c**  $r(f(g(6)))$

6 Consider the following functions:



$$f(x) = -2x - 8$$

$$g(x) = 4x^2 - 4$$

- a Evaluate  $g(f(6))$ .
- b If  $h(x)$  is defined as  $f(g(x))$ , state the equation for  $h(x)$ .
- c What type of functions are  $f(g(x))$  and  $g(f(x))$ ?

7 Consider the following functions:

$$f(x) = -2x + 6$$

$$g(x) = 3x + 1$$

- a If  $r(x)$  is defined as  $f(x^2)$ , state the equation for  $r(x)$ .
- b Hence, state the equation for  $q(x)$ , which is  $g(f(x^2))$ .

8 Consider the following functions:

$$f(x) = x^2 + 3$$

$$g(x) = 4x - 9$$

- a State the equation for  $f(2x)$ .
- b Show that  $f(2x) = g(f(x))$ .

9 The function  $f(x)$  is defined as  $f(x) = -3x + 4g(x)$ .

- a Given that  $f(x)$  is a quadratic function, what type of function is  $g(x)$ ?
- b If  $f(x) = -3x + 20x^2$ , find the equation for  $g(x)$ .
- c Find an algebraic expression for the function  $g(f(x))$ .

10 Consider the following functions:

$$p(x) = x + 3$$

$$q(x) = x^2 - 1$$

$$r(x) = (x + 4)(x + 2)$$

- a Rewrite  $r(x)$  in expanded form.
- b Show that  $r(x) = q(p(x))$ .
- c Write an algebraic expression for  $p(q(x))$ .
- d Find the range of values for which  $r(x) < p(q(x))$ .

11 Consider the following functions:



$$f(x) = \frac{4}{x}$$
$$g(x) = x^2 + 3$$

Find the following functions:

**a**  $f(x)g(x)$

**b**  $f(g(x))$

**c**  $\frac{f(x)}{g(x)}$

**d**  $\frac{g(x)}{f(x)}$

12 Consider the following functions:

$$f(x) = 9x + 4$$
$$g(x) = x^2 - 2x - 3$$

Find the following functions:

**a**  $f(x)g(x)$

**b**  $f(g(x))$

**c**  $\frac{f(x)}{g(x)}$

**d**  $\frac{g(x)}{f(x)}$

## Domain and range

13 For each of the following functions  $f(x)$  and  $g(x)$ :

**i** Find the composite function  $f(g(x))$ .

**ii** State the domain of  $f(g(x))$ .

**iii** Find the composite function  $g(f(x))$ .

**iv** State the domain of  $g(f(x))$ .

**a**  $f(x) = \frac{9}{1-2x}$   
 $g(x) = \frac{3}{x}$

**b**  $f(x) = \sqrt{x+4}$   
 $g(x) = x^2 - 4$

14 Consider the function  $h(x) = \sqrt{1 + \sqrt{1+x}}$  and suppose that  $g(x) = \sqrt{1+x}$ . Find  $f(x)$  such that  $h(x) = f(g(x))$ .

15 Consider the function  $h(x) = \frac{1}{(x-2)^6}$ .

**a** If  $g(x) = x - 2$ , find  $f(x)$  such that  $h(x) = f(g(x))$ .

**b** If  $f(x) = \frac{1}{x}$ , find  $g(x)$  such that  $h(x) = f(g(x))$ .

16 Consider the function  $f(x) = 5x^2 + 4$ . Define  $g(x)$  such that  $g(f(x)) = f(x)$  for all  $x$ .

17 For each of the following functions:



- i State the domain and range of  $f(x)$ .
- ii State the domain and range of  $g(x)$ .
- iii Hence, determine the domain and range of  $f(g(x))$ .

a  $f(x) = x^2$   
 $g(x) = x - 7$

b  $f(x) = \sqrt{x}$   
 $g(x) = x - 3$

c  $f(x) = \sqrt{x}$   
 $g(x) = 4x + 9$

d  $f(x) = x^3$   
 $g(x) = x + 8$

18 Consider the following functions:

$$f(x) = \sqrt{x + 5}$$
$$g(x) = 5x - 6$$

- a Find an algebraic expression for the function  $f(g(x))$ .
- b Express the domain of  $f(g(x))$ , using inequalities.
- c Find an algebraic expression for the function  $g(f(x))$ .
- d Express the domain of  $g(f(x))$ , using inequalities.

19 Consider the following functions:

$$f(x) = -2x$$
$$g(x) = \frac{x}{x - 5}$$

- a Find an algebraic expression for the function  $f(g(x))$ .
- b Determine the value(s) of  $x$  that are not in the domain of  $f(g(x))$ .
- c Find an algebraic expression for the function  $g(f(x))$ .
- d Determine the value(s) of  $x$  that are not in the domain of  $g(f(x))$ .



## Applications

- 20** At sale time in a certain clothing store, all dresses are on sale for \$5 less than 75% of the original price.
- a** Write a function  $g(x)$  that finds 75% of  $x$ .
  - b** Write a function  $f(x)$  that finds 5 less than  $x$ .
  - c** Construct the composite function  $f(g(x))$ .
  - d** Hence, calculate the sale price of a dress that has an original price of \$94.
- 21** A computer manufacturer sells hard drives to a retail outlet, each at a cost of \$11 more than the manufacturing cost. The retail store then sells each hard drive to the public, charging 60% more than they paid to the manufacturer.
- a** If  $m$  represents the manufacturing cost (in dollars), find a function  $A(m)$  which returns the price of buying a hard drive at the retail store.
  - b** Find the price of buying a hard drive at the store, if the manufacturing cost is \$22.50.